FEATURE ENGINEERING

* ML models only understand numbers

$f(x) = y$

mathematical

human

ML

x → features

f

y → label

numerical ??

machine learning

$$X = \begin{bmatrix} x_1 & , & x_2 & , & x_3 & , & x_4 \end{bmatrix}$$

km ✓ | model [categorical] | year ✓ | fuel [petrol, diesel, electric]

Categorical encoding: to convert cat-features into numeric features.

Label encoding

fuel	
P →	0
D →	1
E →	2
H →	3
null →	

P	D	E	H	null
1	0	0	0	
0	1	0	0	
0	0	1	0	
0	0	0	1	

one-hot encoding
dummy variable

weight
23
40
50
60

1000
0100
0010
0001

A	1
B	2
C	3
D	4

→

label encoding

VARIABLE

$$\begin{bmatrix} D \\ E \\ F \end{bmatrix} \qquad \begin{bmatrix} 5 \\ 6 \end{bmatrix} \checkmark$$

ORDINAL

ordered categories

$A > B > C > \dots > F$

price

$$\begin{bmatrix} [\ ] \\ [\ ] \\ [\ ] \end{bmatrix}$$

S → 0
M → 1
L → 2

car - price

	km	old [0-10]
5,000	20,000	1

200,000

50,000

80,000

5

3

4

Scaling

not to use on outlier data

100,000

Standard scaling

mean = m

S.D :

20,000

mean

Min Max Scaler : Normalization

X_1		X_2	
4000	0	30	0
⋮	⋮	⋮	⋮
40,000	1	80	1
	(0-1)		(0-1)

$$\frac{x - \min}{\max - \min}$$

Robust Scaler